Name : Rohita M
Roll no : 241501168

Exp No: 4	Setup and configure a LAN using a Switch and Ethernet cables

AIM:

To setup and configure a LAN using a switch and Ethernet cables, assign IP addresses to the hosts, check the connectivity using the ping commandand share files and folders across the machines of the LAN.

ALGORITHM / PROCEDURE:

1. Connect 4 hostmachines totheswitch using straight-through UTP cables (RJ45). Check that the link LEDs of the switch glow.
2. Assign a static IP address to each host: **Windows**, Control Panel > Network and Sharing Centre > Change adapter settings > Ethernet > Properties > IPv4 (or run ncpa.cpl).
3. Use the addresses 192.168.10.11 to 192.168.10.14 with subnet mask 255.255.255.0. Leave the default gateway empty because there is only one network.
4. Open the Command Prompt and verify the settings with ipconfig.
5. Check the connectivity between the machines with ping, for example from PC1 ping `192.168.10.12`.
6. On PC1 create a folder LabShare, right click > Properties > Sharing > Advanced Sharing > tick **Share this folder** and give **Read/Write** permission to **Everyone**.
7. On PC2 open **File Explorer** and type \\192.168.10.11 in the address bar. The files of PC1 are displayed.
8. The folder can also be mapped as a drive with net use Z: \\192.168.10.11\LabShare and the shared files can be copied between the machines.
9. Note down the output of eachcommand.

Host	IP address	Subnet mask	Switch port
PC1	`192.168.10.11`	`255.255.255.0`	Fa0/1
PC2	`192.168.10.12`	`255.255.255.0`	Fa0/2
PC3	`192.168.10.13`	`255.255.255.0`	Fa0/3
PC4	`192.168.10.14`	`255.255.255.0`	Fa0/4

DIAGRAM:

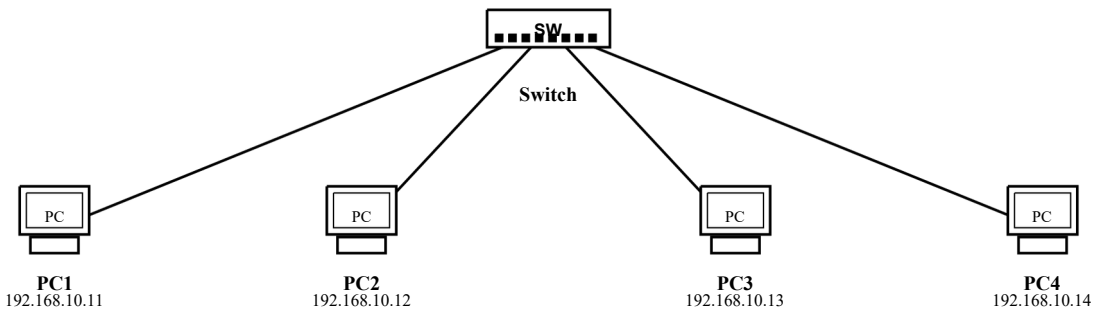


Fig 1: LAN with 4 hosts connected to a switch (star topology)

OUTPUT:

CommandPrompt - PC1

```
C:\Users\userUSER >ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix . :
   Link-local IPv6 Address . . . . . : fe80::6c1f:3a2e:91b4:d7c2%8
   IPv4 Address. . . . . . . . . . . : 192.168.10.11
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
Default Gateway . . . . . . . . . :
```

Command Prompt - PC1

```
C:\Users\userUSER >ping 192.168.10.12

Pinging 192.168.10.12 with 32 bytes of data:
Reply from 192.168.10.12: bytes=32 time=1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.12:Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:

    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\userUSER >ping 192.168.10.13

Pinging 192.168.10.13 with 32 bytes of data:
Reply from 192.168.10.13: bytes=32 time<1ms TTL=128
Reply from 192.168.10.13: bytes=32 time<1ms TTL=128
Reply from 192.168.10.13: bytes=32 time=1ms TTL=128
Reply from 192.168.10.13: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.13:

    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\userUSER >ping 192.168.10.14

Pinging 192.168.10.14 with 32 bytes of data:
Reply from 192.168.10.14: bytes=32 time<1ms TTL=128
Reply from 192.168.10.14: bytes=32 time<1ms TTL=128
Reply from 192.168.10.14: bytes=32 time<1ms TTL=128
Reply from 192.168.10.14: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

Command Prompt - PC2

```
C:\Users\userUSER >net view \\192.168.10.11
Shared resources at \\192.168.10.11

Share name   Type   Used as   Comment
-------------------------------------------------------
LabShare     Disk The command
completed successfully.
C:\Users\userUSER >net use Z: \\192.168.10.11\LabShare
The command completed successfully.


C:\Users\userUSER >dir Z:
 Volume in drive Z is LabShare
 Directory of Z:\ 07-10-2026 08:42 PM
<DIR>          .
```

Command Prompt - PC2 (contd.)

```
07-10-2026  08:42  PM  <DIR>          ..
07-10-2026  08:40  PM          14,336 CN_Lab_Notes.docx
07-10-2026  08:41  PM         212,480 Exp4_Topology.png
07-10-2026 08:42 PM             1,024 hosts_list.txt
               3 File(s)        227,840 bytes
               2 Dir(s) 48,213,327,872 bytes free

C:\Users\userUSER >copy Z:\hosts_list.txt C:\Users\userUSER
       \Desktop         1 file(s) copied.
```

RESULT:

ThusaLANwas set up with 4 hosts connected to a switch using Ethernet cables, the IP addresses were assigned, theconnectivity was verified using the ping command and files were shared and accessed across the machinesof the LAN.